Introduction to Databases

Lecture 01

Borough of Manhattan Community College

Chapter Objectives

In this chapter you will learn:

- The importance of database systems.
- Some common uses of database systems.
- The characteristics of file-based systems.
- The problems with the file-based approach.
- The meaning of the term "database."
- The meaning of the term "database management system" (DBMS).
- The typical functions of a DBMS.
- The major components of the DBMS environment.
- The personnel involved in the DBMS environment.
- The advantages and disadvantages of DBMSs.

Introduction

- A **database (DB)** is a collection of related data.

- A **database management system (DBMS)** is the software that manages and controls access to the DB.

- A **database application** is simply a program that interacts with the DB at some point in its execution.

- A **database system** is a collection of application programs that interact with the DB and the DBMS.

Supermarket DB Example

- A bar code reader scans each of your purchases.
- An application uses the bar code to find out the price of the item from a product DB.
- The application displays the price on the cash register and reduces the number of such items in stock.
- If the reorder level falls below a threshold, the database system places an order to obtain more of that item.
- Assistants can check whether an item is in stock by running an application program that determines availability from the DB.

Library DB Example

- A library has a DB containing details of the books in the library, details of the readers, reservations …
- An application allows readers to find a book based on its title, authors, or subject area.
- The database system handles reservations to allow a reader to reserve a book and to be informed by email when the book is available.
- The system sends reminders to borrowers who have failed to return books by the due date.

Other Examples

- Booking a vacation
- Buying products online
- Being hospitalized
- Studying at college
- …

File-Based System

- The **file-based system** is a collection of application programs that perform services for the end-users, such as the production of reports. Each program defines and manages its own data.

- File-based systems were an early attempt to computerize the manual filing system that we are all familiar with.

- In our own homes, we probably have some sort of filing system that contains receipts, warranties, bills, bank statements.

File-Based System

- Understanding the problems in file-based systems may prevent us from repeating these problems in database systems.

- If you wish to convert a file-based system to a database system, understanding how the file-based system works is extremely useful.

- Example of a file-based system:

  A typical real estate agent's office might have a separate file for each property for rent, each potential renter, and each member of the staff.

File-Based System - DreamHome Example

- The Sales Department is responsible for the selling and renting of properties. Whenever an owner wishes to offer his or her property as a rental, he or she approaches the Sales Department and fills the **Property for Rent Details Form**.

- The Sales Department also handles inquiries from clients, and a **Client Details Form** is completed for each one.

- The Sales Department creates an information system to handle the renting of property. The system consists of three files containing property, owner, and client details.

Sales Department
Property for Rent Details form

DreamHome
Property for Rent Details
Property Number: PG21

Address 18 Dale Rd

City Glasgow

Postcode G12

Type House **Rent** 600

No. of Rooms 5

Allocated to Branch:
163 Main St, Glasgow

Branch No. B003

Staff Responsible
Ann Beech

Owner's Details

Name Carol Farrel

Address 6 Achray St
Glasgow G32 9DX

Tel. No. 0141-357-7419

Owner No. C087

Business Name ________

Address ________
____

Tel. No. ________

Owner No. ________

Contact Name ________

Business Type ________

Sales Department
Client Details form

DreamHome
Client Details
Client Number: CR74

First Name Mike
Last Name Ritchie
Address 18 Tain St
PA1G 1YQ
Tel. No. 01475-392178

Property Requirement Details

Preferred Property Type House
Maximum Monthly Rent 750
General Comments Currently living at home with parents
Getting married in August

Seen By Ann Beech
Date 24-Mar-13
Branch No. B003
Branch City Glasgow

PropertyForRent

propertyNo	street	city	postcode	type	rooms	rent	ownerNo
PA14	16 Holhead Rd	Aberdeen	AB7 5SU	House	6	650	CO46
PL94	6 Argyll St	London	NW2	Flat	4	400	CO87
PG4	6 Lawrence St	Glasgow	G11 9QX	Flat	3	350	CO40
PG36	2 Manor Rd	Glasgow	G32 4QX	Flat	3	375	CO93
PG21	18 Dale Rd	Glasgow	G12	House	5	600	CO87
PG16	5 Novar Dr	Glasgow	G12 9AX	Flat	4	450	CO93

PrivateOwner

ownerNo	fName	lName	address	telNo
CO46	Joe	Keogh	2 Fergus Dr, Aberdeen AB2 7SX	01224-861212
CO87	Carol	Farrel	6 Achray St, Glasgow G32 9DX	0141-357-7419
CO40	Tina	Murphy	63 Well St, Glasgow G42	0141-943-1728
CO93	Tony	Shaw	12 Park Pl, Glasgow G4 0QR	0141-225-7025

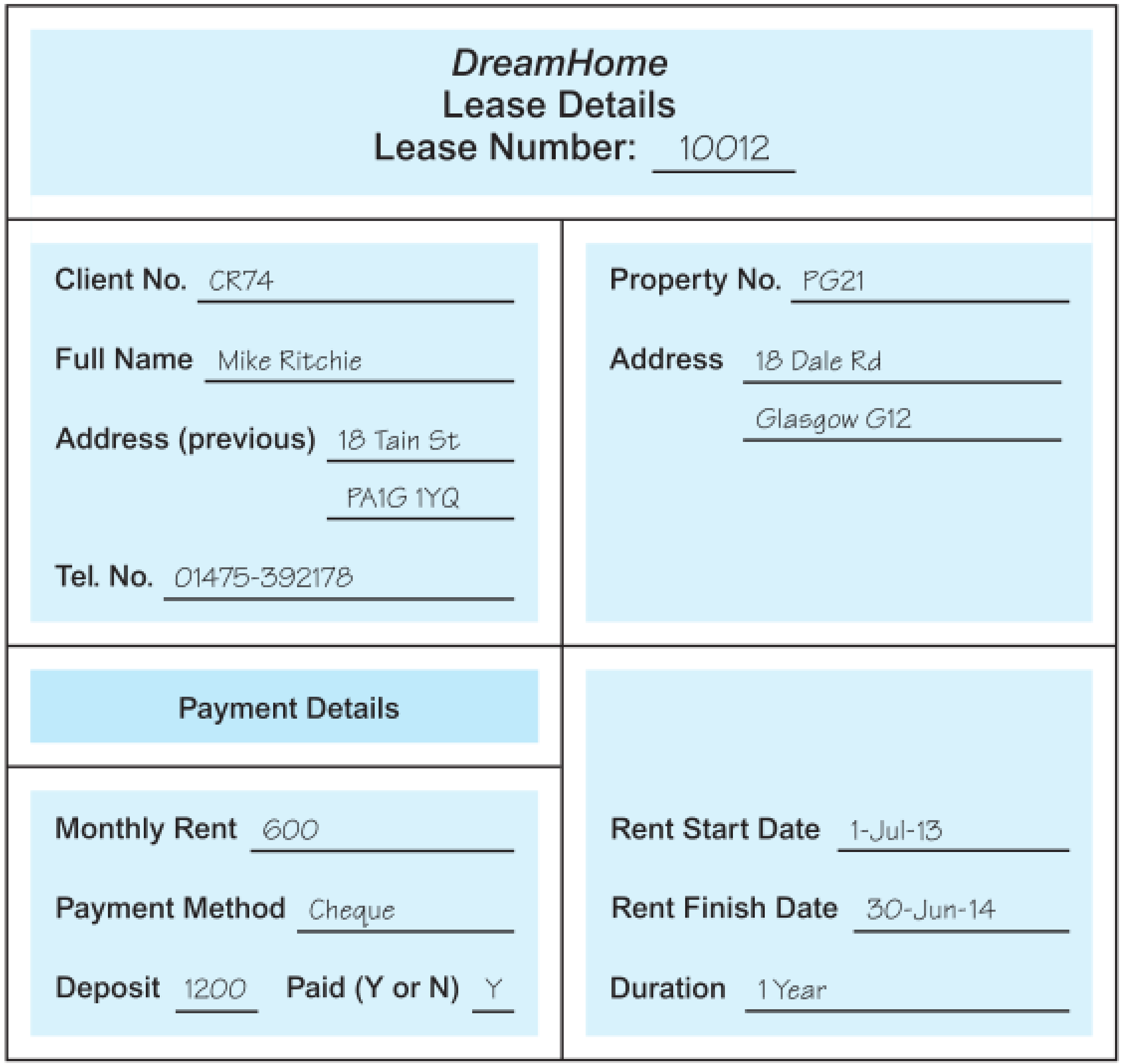
Client

clientNo	fName	lName	address	telNo	prefType	maxRent
CR76	John	Kay	56 High St, London SW1 4EH	0207-774-5632	Flat	425
CR56	Aline	Stewart	64 Fern Dr, Glasgow G42 0BL	0141-848-1825	Flat	350
CR74	Mike	Ritchie	18 Tain St, PA1G 1YQ	01475-392178	House	750
CR62	Mary	Tregear	5 Tarbot Rd, Aberdeen AB9 3ST	01224-196720	Flat	600

Figure 1.2 The PropertyForRent, PrivateOwner, and Client files used by Sales.

- Whenever a client agrees to rent a property, a form with the client and property details is filled and passed to the Contracts Department.

Lease Details form used by Contracts Department.



DreamHome
Lease Details
Lease Number: 10012

Client No. CR74
Full Name Mike Ritchie
Address (previous) 18 Tain St
PA1G 1YQ
Tel. No. 01475-392178

Property No. PG21
Address 18 Dale Rd
Glasgow G12

Payment Details

Monthly Rent 600
Payment Method Cheque
Deposit 1200 **Paid (Y or N)** Y

Rent Start Date 1-Jul-13
Rent Finish Date 30-Jun-14
Duration 1 Year

- The Contracts Department creates an information system to handle lease agreements. The system consists of three files that store lease, property, and client details.

Lease

leaseNo	propertyNo	clientNo	rent	payment Method	deposit	paid	rentStart	rentFinish	duration
10024	PA14	CR62	650	Visa	1300	Y	1-Jun-13	31-May-14	12
10075	PL94	CR76	400	Cash	800	N	1-Aug-13	31-Jan-14	6
10012	PG21	CR74	600	Cheque	1200	Y	1-Jul-13	30-Jun-14	12

PropertyForRent

propertyNo	street	city	postcode	rent
PA14	16 Holhead	Aberdeen	AB7 5SU	650
PL94	6 Argyll St	London	NW2	400
PG21	18 Dale Rd	Glasgow	G12	600

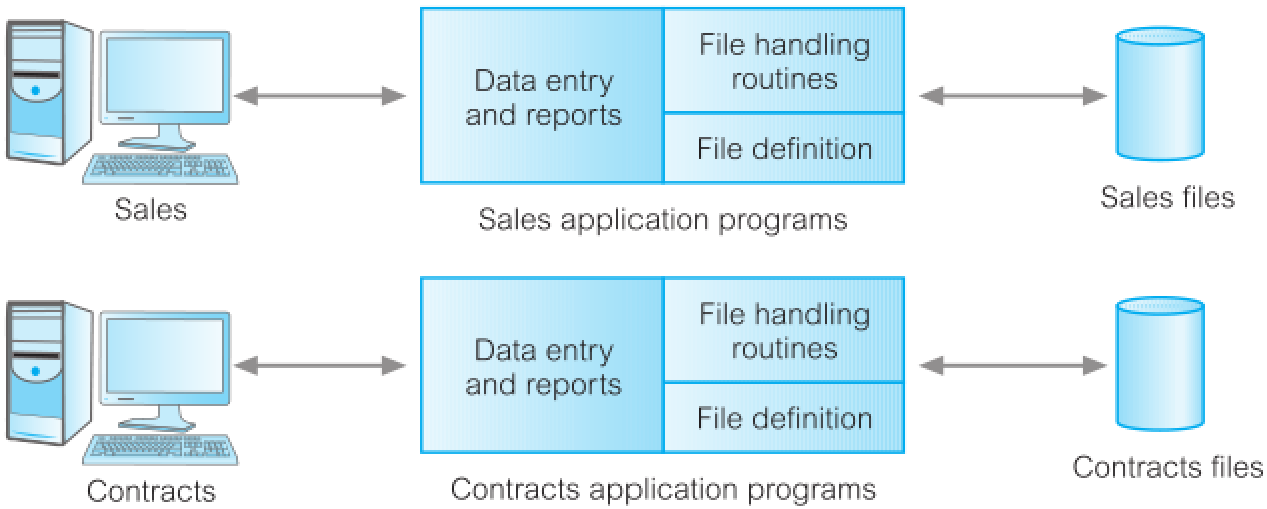
Client

clientNo	fName	lName	address	telNo
CR76	John	Kay	56 High St, London SW1 4EH	0171-774-5632
CR74	Mike	Ritchie	18 Tain St, PA1G 1YQ	01475-392178
CR62	Mary	Tregear	5 Tarbot Rd, Aberdeen AB9 3ST	01224-196720

Figure 1.4 The Lease, PropertyForRent, and Client files used by the Contracts Department.

- Each department accessing their own files through application programs written especially for them.

- Each set of departmental application programs handles data entry, file maintenance, and the generation of a fixed set of specific reports.



Sales Files

PropertyForRent (propertyNo, street, city, postcode, type, rooms, rent, ownerNo)
PrivateOwner (ownerNo, fName, lName, address, telNo)
Client (clientNo, fName, lName, address, telNo, prefType, maxRent)

Contracts Files

Lease (leaseNo, propertyNo, clientNo, rent, paymentMethod, deposit, paid, rentStart, rentFinish, duration)
PropertyForRent (propertyNo, street, city, postcode, rent)
Client (clientNo, fName, lName, address, telNo)

File-Based System

- We can find similar examples in other departments:

  - The Payroll Department stores details relating to each member of staff's salary:

    StaffSalary (staffNo, fName, lName, sex, salary, branchNo)

  - The Human Resources (*HR*) Department also stores staff details:

    Staff (staffNo, fName, lName, position, sex, dateOfBirth, salary, branchNo)

- Significant amount of duplication of data in these departments, and this is generally true of file-based systems.

File-Based System – Limitations

- Separation and isolation of data.
- Duplication of data.
- Data dependence.
- Incompatible file formats.
- Fixed queries/proliferation of application programs.

File-Based System – Limitation #1

Separation and isolation of data

- When data is isolated in separate files, it is more difficult to access data that should be available.

- For example, if we want to produce a list of all houses that match the requirements of clients:

  - We first need to create a temporary file of those clients who have "house" as the preferred type.

  - For each client in the temporary file, we search the PropertyForRent file for those properties with type "house" and the rent is less than the client's maximum rent.

  - The application developer must synchronize the processing of two files to ensure that the correct data is extracted.

  - This difficulty is compounded if we require data from more than two files.

File-Based System – Limitation #2

Duplication of data

- Due to the decentralized approach taken by each department, the file-based approach necessitated the uncontrolled duplication of data.
- In the DreamHome example we can clearly see that there is duplication of both property and client details in the Sales and Contracts Departments.
- Uncontrolled duplication of data is undesirable for several reasons, including:
  - Duplication is wasteful. It costs time and money to enter the data.
  - It takes up additional storage space, again with associated costs.
  - More importantly, duplication can lead to loss of data integrity:

    - A paycheck can be sent to the wrong address.

    - An employee can receive the wrong salary.

File-Based System – Limitation #3

Data dependence

- The physical structure and storage of the data files and records are defined in the application code.

- Changes to an existing structure are difficult to make.

  For example, converting the *rent* field in the *PropertyForRent* from integer to float requires the creation of a one-off program:
  - Open the original *PropertyForRent* file for reading.
  - Open a temporary file with the new structure.
  - Read a record from the original file, convert the data to conform to the new structure, and write it to the temporary file, then repeat this step for all records.
  - Delete the original *PropertyForRent* file.
  - Rename the temporary file as *PropertyForRent*.

- In addition, all programs that access the *PropertyForRent* file must be modified to conform to the new file structure. Clearly, this process could be very time-consuming and subject to error.

File-Based System – Limitation #4

Incompatible file formats

- Because the structure of files is embedded in the application programs, the structures are dependent on the application programming language.
- For example, the structure of a file generated by a *COBOL* program may be different from the structure of a file generated by a *C* program.
- The direct incompatibility of such files makes them difficult to process jointly.
- Therefore, cross-referencing files requires that an application developer write software to convert the files to some common format to facilitate processing.
- Again, this process can be time-consuming and expensive.

File-Based System – Limitation #5

Fixed queries/proliferation of application programs

- File-based systems are very dependent upon the application developer, who has to write any queries or reports that are required.
- In some organizations, the type of query or report that could be produced was fixed. There was no facility for asking unplanned queries.
- In other organizations, there was a proliferation of files and application programs. Very difficult for Data Processing Department to handle all the work, and programs are inefficient in meeting the demands of the users:
  - Limited documentation.
  - Difficult maintenance.
  - No provision for security or integrity.
  - Limited recovery in the event of a hardware or software failure.
  - Access to the files restricted to one user at a time.

Database Approach

- A **database** is a shared collection of logically related data and its description.

- The database is no longer owned by one department but is a shared corporate resource. It can be used simultaneously by many departments and users.

- The database holds not only the organization's operational data, but also a description of this data (system catalog or data dictionary).

Database Approach

- The database approach separates the structure of the data from the application programs and stores it in the database.

- If new data structures are added or existing structures are modified, then the application programs are unaffected, provided that they do not directly depend upon what has been modified.

- For example, if we add a new field to a file, existing applications are unaffected. However, if we remove a field from a file that an application program uses, then that application program is affected by this change and must be modified accordingly.

The Database

- When we analyze the information needs of an organization, we attempt to identify entities, attributes, and relationships.

  - An **entity** is a distinct object in the organization that is to be represented in the database (a person, place, thing, concept, or event).

  - An **attribute** is a property that describes some aspect of the object that we wish to record (ex: *fName*, *lName*, *address*, *salary*, *gpa*, …).

  - A **relationship** is an association between entities.

- The database represents the entities, the attributes, and the logical relationships between the entities.

The Database Management System (DBMS)

- A **DBMS** is the software that manages and controls access to the DB.

- Typically, a DBMS provides the following facilities:

  - It allows users to define the database, usually through a Data Definition Language (*DDL*).

  - It allows users to insert, update, delete, and retrieve data from the database, usually through a Data Manipulation Language (*DML*).

  - It provides controlled access to the database. For example, it may provide:

    - A security system, which prevents unauthorized users accessing the database.

    - An integrity system, which maintains the consistency of stored data.

    - Concurrency control system, which allows shared access of the database.

    - Recovery control system, which restores the database to a previous state.

Application Programs

- A database application is a computer program that interacts with the database by issuing an appropriate request to the DBMS.

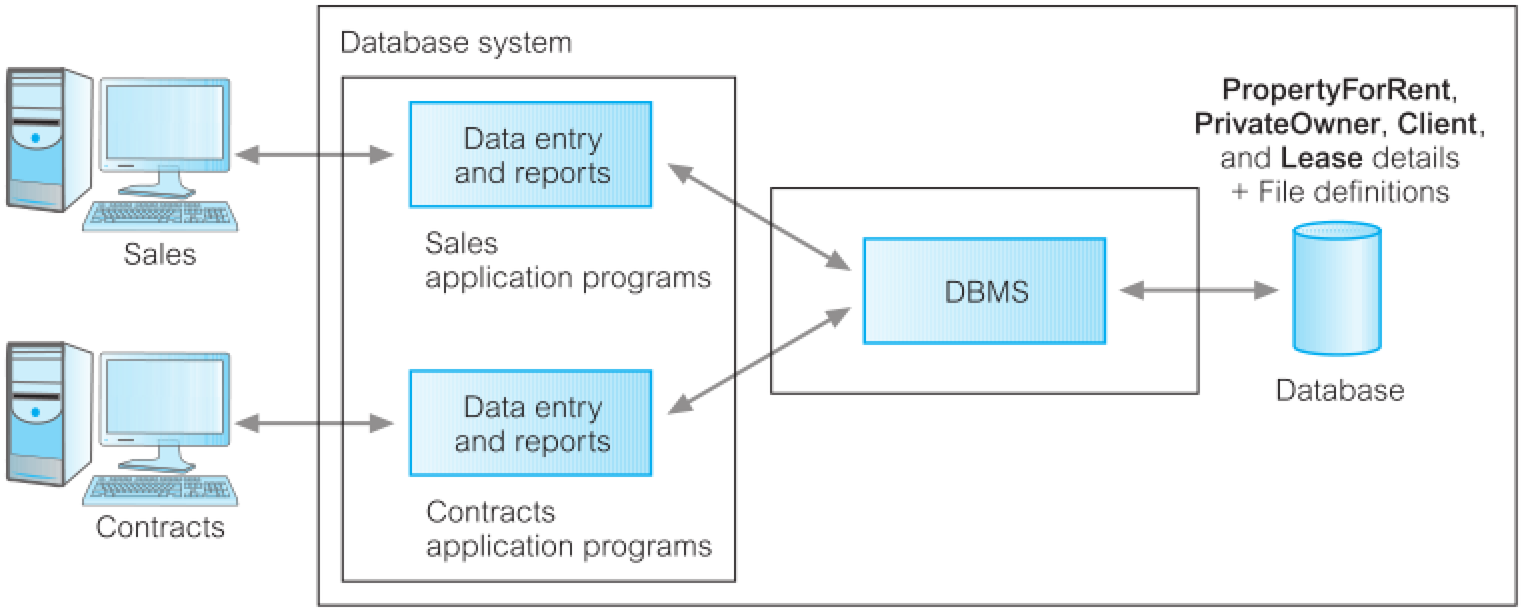
- Users interact with the database through a number of application programs that are used to create and maintain the database and to generate information.

  - The Sales and Contracts Departments using their application programs to access the database through the DBMS.

  - Each set of departmental application programs handles data entry, data maintenance, and the generation of reports.

- However, as opposed to the file-based approach, the physical structure and storage of the data are now managed by the DBMS.

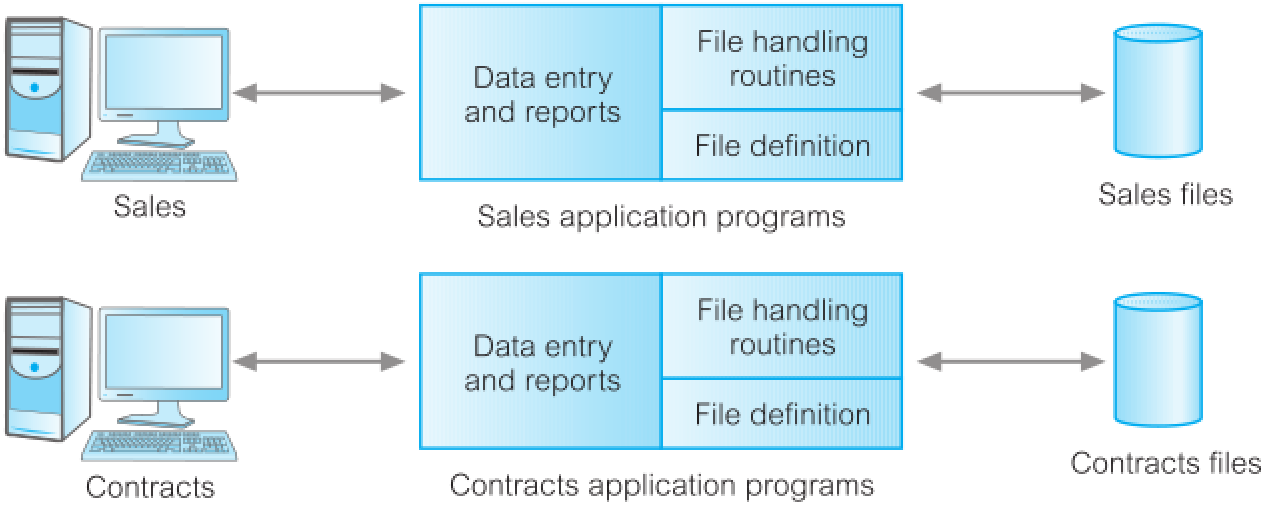
Database Approach



PropertyForRent (propertyNo, street, city, postcode, type, rooms, rent, ownerNo)
PrivateOwner (ownerNo, fName, lName, address, telNo)
Client (clientNo, fName, lName, address, telNo, prefType, maxRent)
Lease (leaseNo, propertyNo, clientNo, paymentMethod, deposit, paid, rentStart, rentFinish)

File-Based Approach

Reminder



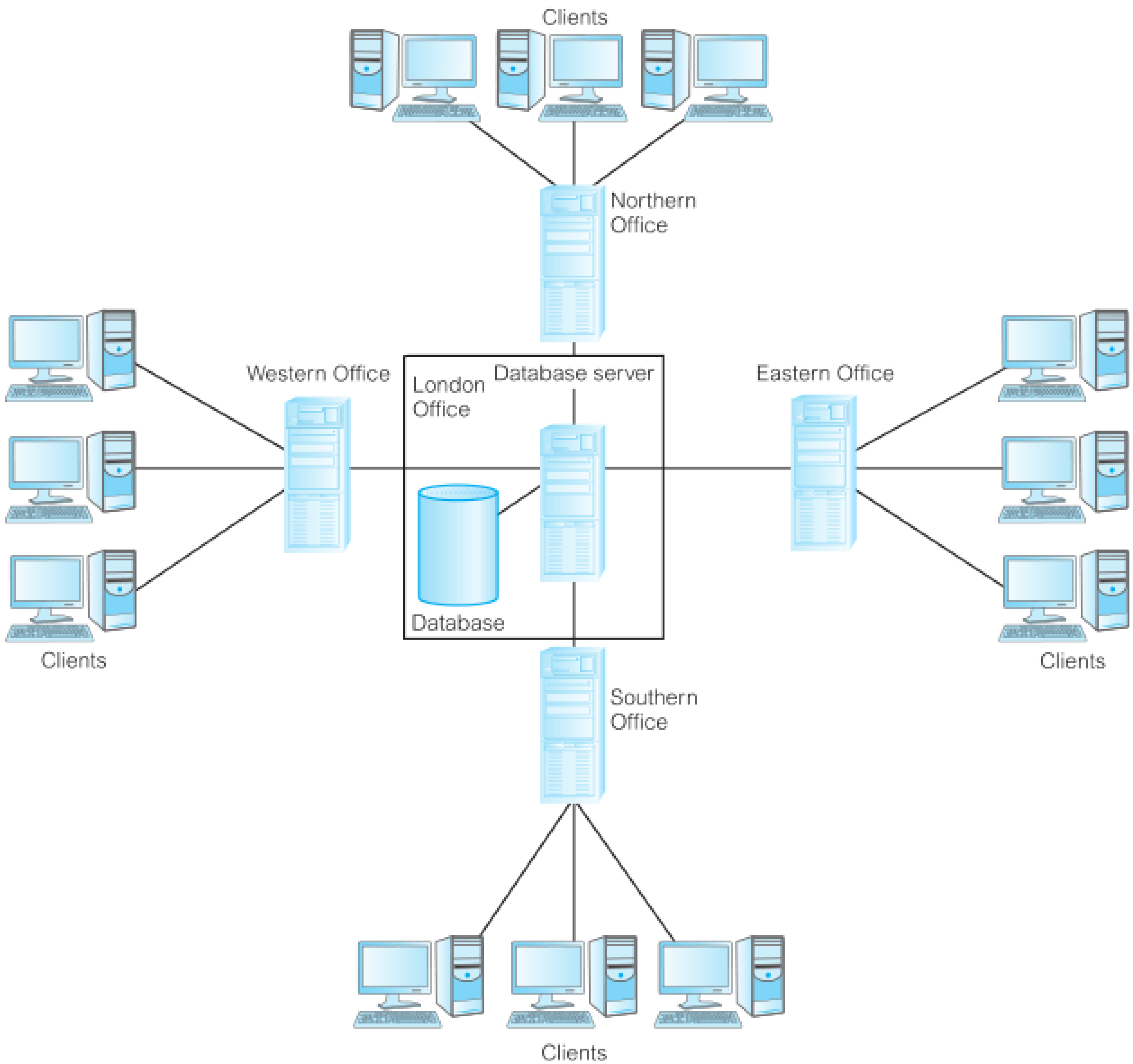
Sales Files

PropertyForRent (propertyNo, street, city, postcode, type, rooms, rent, ownerNo)
PrivateOwner (ownerNo, fName, lName, address, telNo)
Client (clientNo, fName, lName, address, telNo, prefType, maxRent)

Contracts Files

Lease (leaseNo, propertyNo, clientNo, rent, paymentMethod, deposit, paid, rentStart, rentFinish, duration)
PropertyForRent (propertyNo, street, city, postcode, rent)
Client (clientNo, fName, lName, address, telNo)

DreamHome hardware configuration.



Components of the DBMS Environment

- We can identify five major components in the DBMS environment:

1. **Hardware**: The hardware can range from a single personal computer to mainframes or a network of computers.

2. **Software**: The software component comprises the DBMS software itself and the application programs, together with the operating system, including network software if the DBMS is being used over a network.

3. **Data**: The database contains both the operational data and its description.

4. **Procedures**: Procedures refer to the instructions and rules that govern the design and use of the database.

5. **People**: The final component is the people involved with the system.

Roles in the Database Environment

- We can identify four distinct types of people in the DBMS environment:

1. Data and Database Administrators (***DBA***): Responsible for the management of the data resource and the physical realization of the database.

2. Database Designers: They are concerned with identifying the data (that is, the entities and attributes), the relationships between the data, and the constraints on the data.

3. Application Developers: Once the database has been implemented, the application programs that provide the required functionality for the end-users must be implemented. This is the responsibility of the application developers.

4. End-Users: They access the database through specially written application programs that attempt to make the operations as simple as possible.

Advantages of DBMSs

- **Control of data redundancy:** The database approach attempts to eliminate the redundancy by integrating the files so that multiple copies of the same data are not stored.

- **Data consistency:** By eliminating or controlling redundancy, we reduce the risk of inconsistencies occurring.
  - If a data item is stored only once in the database, any update to its value has to be performed only once and the new value is available immediately to all users.

- **Sharing of data:** The database belongs to the entire organization and can be shared by all authorized users.

Advantages of DBMSs

- **Improved data integrity:** Database integrity refers to the validity and consistency of stored data.
  - Integrity is usually expressed in terms of constraints, which are consistency rules that the database is not permitted to violate.
  - DBA defines integrity constraints, and the DBMS enforces them.

- **Improved security:** Database security is the protection of the database from unauthorized users.
  - This security may take the form of user names and passwords to identify people authorized to use the database.

Advantages of DBMSs

- **Increased productivity:** The DBMS provides all the low-level file-handling routines that are typical in application programs.
  - The provision of these functions allows the programmer to concentrate on the specific functionality required by the users without having to worry about low-level implementation details.
- **Increased concurrency:** In some file-based systems, if two or more users are allowed to access the same file simultaneously, it is possible that the accesses will interfere with each other.
  - DBMSs manage concurrent database access and ensure that such problems cannot occur.
- **Improved backup and recovery services:** Modern DBMSs provide facilities to minimize the amount of processing to protect the data from failures to the computer system or application program.

Disadvantages of DBMSs

- **Complexity:** DBMS is an extremely complex piece of software.
  - Database designers and developers, data and database administrators, and end-users must understand this functionality to take full advantage of it.

- **Cost of DBMSs:** The cost of DBMSs varies significantly.
  - A single-user DBMS for a personal computer may only cost $100. However, a large mainframe multi-user DBMS servicing hundreds of users can be extremely expensive, perhaps $100,000 or even $1,000,000.
  - There is also the recurrent annual maintenance cost.

Disadvantages of DBMSs

- **Cost of conversion:** The cost of converting existing applications to run on the new DBMS and hardware is very expensive.
  - This also includes the cost of training staff to use these new systems.
  - This cost is one of the main reasons why some organizations feel tied to their current systems and cannot switch to more modern database technology.

- **Greater impact of a failure:** The centralization of resources increases the vulnerability of the system.
  - All users and applications rely on the availability of the DBMS, the failure of certain components can bring operations to a halt.